

PMM3 vs MLE-TN: Comparison of Semiparametric and Parametric Estimation of Linear Regression Parameters under Two-piece Normal Distributed Errors

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Abstract. This paper addresses the problem of estimating the parameter vector of a linear regression model under symmetric non-Gaussian errors following a Two-piece Normal (TN) distribution. Three methods are compared: Ordinary Least Squares (OLS), Maximum Likelihood Estimation for the TN distribution (MLE-TN), and the third-power Polynomial Maximization Method (PMM3). MLE-TN is fully parametric; PMM3 is semiparametric, using only the 2nd, 4th and 6th-order moments. Theoretical analysis shows that PMM3 achieves up to a 95.5% variance reduction compared to OLS at $\lambda = 5$ (the TN shape parameter), while MLE-TN provides up to a 28-fold advantage for large λ . A Monte Carlo study confirmed that the empirical efficiency coefficient g_3 converges to the theoretical value for $n \geq 100$. Under misspecification PMM3 never falls below OLS (ARE from 1.02 to 2.68), whereas MLE-TN can be less efficient than OLS. On real-world data (*Iris versicolor*), PMM3 achieves a 43% bootstrap variance reduction and improves LOO-CV by 2.3% over OLS. An empirical efficiency threshold for PMM3 is established at $\gamma_4 \lesssim -0.7$.

Keywords: Polynomial Maximization Method, Linear Regression, Two-piece Normal distribution, Maximum Likelihood Estimation, Estimation Efficiency, Robustness, Monte Carlo.

1 Introduction

The Ordinary Least Squares (OLS) method [5] is the Best Linear Unbiased Estimator under Gaussian errors, but its efficiency depends on the shape of the error distribution [2]. When normality is violated, OLS loses its asymptotic optimality, whereas estimators exploiting higher-order moments may achieve lower variance.

In practical applications, including metrology, medicine, econometrics, and the natural sciences, measurement errors frequently follow symmetric but non-Gaussian distributions. Platykurtic distributions ($\gamma_4 < 0$) commonly arise in measurements with

a limited operating range, in regression residual histograms, and when multiple sources of uniformly distributed errors are aggregated.

One important family of such distributions is the Two-piece Normal (TN) distribution, introduced in [8] and further investigated in [14]. The TN distribution covers a broad spectrum of symmetric shapes, ranging from an approximately Gaussian form at ($\lambda \approx 0$) to a strongly bimodal form at ($\lambda \gg 1$), making it a natural benchmark for studying symmetric platykurtic distributions with $\gamma_4 \in (-2, 0)$.

Three fundamentally different approaches can be distinguished: Parametric approach (MLE-TN): jointly estimates all parameters (β, η, λ) , attains the Cramér–Rao lower bound under correct model specification, but is sensitive to model misspecification [6], [18]; Semiparametric approach (PMM3): utilizes only the 2-nd, 4-th and 6-th order moments. It does not require knowledge of the functional form of the error distribution and remains robust under model misspecification; Nonparametric and robust approaches: including Weighted Least Squares (WLS), Least Absolute Deviations (LAD) regression [13], quantile regression [9], and adaptive Maximum Likelihood Estimation [15].

The present study addresses the following practical question: under what conditions, and to what extent, does PMM3 outperform both MLE-TN and OLS.

1.1 Literature Review

In [2], a general expression for the asymptotic relative efficiency of OLS was derived in terms of the cumulants of the underlying distribution. Subsequently, [16] systematically investigated parameter estimation in linear regression models with symmetric non-Gaussian errors. The studies reported in [1] and [4] confirmed that, for $\gamma_4 < -0.5$, classical OLS may perform worse than the optimal estimator. Robust regression techniques [6] and LAD regression [13] are primarily designed for leptokurtic ($\gamma_4 > 0$) distributions. In contrast, for platykurtic ($\gamma_4 < 0$) distributions, none of the standard methods can be regarded as optimal. In [12], a semiparametric regression framework with an unspecified error distribution was developed and shown to outperform MLE when the assumption of normality is misspecified. In [14], both Bayesian and frequentist approaches to linear regression with TN-distributed errors were proposed, and MLE estimation was investigated using L-BFGS-B optimization. Furthermore, [11] developed an objective Bayesian methodology for the broader class of two-piece distributions. Both studies emphasize that, under model misspecification, parametric approaches may be inferior to semiparametric alternatives [18].

As for the Polynomial Maximization Method (PMM), it was developed by Kunchenko and Lega [10]. The theoretical efficiency coefficient of PMM3 derived in [17] is given in Section 2.4.

In [20], PMM was extended to linear regression with asymmetric errors (PMM2, $g_2 = 1 - \gamma_3^2 / (2 + \gamma_4)$) and in [19], the first application of the PMM3 to linear regression with symmetric non-Gaussian errors of general form was presented. In [7], a

normality test based on the sample skewness and kurtosis was proposed. The corresponding adaptive decision algorithm is: normally distributed residuals – apply OLS; asymmetric residuals – apply PMM2; symmetric platykurtic residuals – apply PMM3.

1.2 Scientific contribution of the study

Although previous studies have investigated PMM3 for general symmetric distributions and MLE-TN for the Two-piece Normal model, a systematic direct comparison between PMM3 and MLE-TN within the TN framework has not previously been conducted. The principal contributions of this work are as follows:

(i) Theoretical comparison: derivation and analysis of analytical expressions for $g_3(\lambda)$, $ARE(\text{MLE-TN})(\lambda)$, and the efficiency curve comparing PMM3 and MLE-TN; (ii) Monte Carlo validation: $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$, $n \in \{46, 100, 200, 500\}$, using $M = 500$ replications; (iii) Robustness analysis: demonstration that PMM3 does not fall below OLS under model misspecification (ARE from 1.02 to 2.68), whereas MLE-TN may experience a substantial loss of efficiency; (iv) Numerical stability: verification that the Newton-Raphson algorithm converges in 100% of the replications across all 24 Monte Carlo scenarios; (v) Empirical threshold $\gamma_4 \lesssim -0.7$: establishment and justification of an empirical efficiency threshold based on seven real-world datasets; (vi) Real-data application: analysis of the Iris versicolor dataset ($n = 50$, $\gamma_4 = -0.966$), demonstrating that PMM3 achieves a 43% reduction in variance.

2 Materials and methods

2.1 Linear regression model

Consider the following linear regression model:

$$y_v = x_v^T \beta + \varepsilon_v, \quad v = 1, \dots, n, \quad (1)$$

where y_i denotes the response variable, $x_i \in \mathbb{R}^p$ is the vector of regressors, $\beta \in \mathbb{R}^p$ is the vector of unknown regression parameters, and ε_i are independent and identically distributed random errors.

Assumption. For PMM3: the error term satisfies $\varepsilon_v \sim$ a symmetric distribution with $E(\varepsilon) = 0$ and possesses finite moments up to the sixth order. **For MLE-TN:** the error term follows the Two-piece Normal distribution, $\varepsilon_v \sim TN(0, \eta, \lambda)$.

2.2 Two-piece Normal (TN) distribution

The Two-piece Normal (TN) distribution with zero location parameter, scale parameter $\eta > 0$, and shape parameter $\lambda > 0$ has the probability density function [14]:

$$f(z | \lambda) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{z^2}{2} - \frac{\lambda^2}{2}\right) \cdot \cosh(\lambda z), \quad z \in \mathbb{R}. \quad (1)$$

The principal properties of the TN distribution are as follows: $\lambda \rightarrow 0$ – the distribution approaches the standard normal distribution $N(0, 1)$; $\lambda > 1$ – the distribution becomes bimodal, exhibiting two symmetric modes; γ_4 is strictly negative for all values of λ , indicating a platykurtic distribution.

The normalized excess kurtosis is given by:

$$\gamma_4 = \frac{\mu_4}{\mu_2^2} - 3 < 0 \quad \text{for all } \lambda > 0. \quad (1)$$

2.3 MLE-TN: Maximum likelihood estimation

The MLE-TN approach jointly estimates the parameter vector (β, η, λ) by maximizing the log-likelihood function:

$$l(\beta, \eta, \lambda). \quad (1)$$

The optimization procedure is performed using the L-BFGS-B algorithm with the following initial values: $\beta_0 = \hat{\beta}_{OLS}$, $\eta_0 = \exp(\log SD(\hat{e}_{OLS}))$, $\lambda_0 = 1.0$. Advantages: achieves the Cramér–Rao lower bound under correct model specification; provides $ARE(MLE - TN / OLS) \gg 1$ for sufficiently large values of λ . Limitations: requires estimation of $p + 2$ parameters; is sensitive to model misspecification; may exhibit numerical instability for small sample sizes and large n values of λ .

2.4 PMM3: Polynomial Maximization Method (S = 3)

The PMM3 estimator evaluates only the regression parameter vector β by employing a moment-corrected score function. The algorithm consists of the following steps: compute the initial estimate $\beta^{(0)} = \hat{\beta}_{OLS}$ and evaluate the sample central moments of the residuals m_2, m_4, m_6 , then the correction coefficient is estimated once according to

$$\kappa = \frac{m_6 - 3 m_4 m_2}{m_4 - 3 m_2^2}. \quad (1)$$

If the corresponding condition is violated $|m_4 - 3 m_2^2| < 10^{-10}$, the algorithm returns the Gaussian fallback solution $\hat{\beta}_{OLS}$.

The parameter vector is updated according to the Newton-Raphson procedure: $Z_{1v} = \varepsilon_v(\kappa - \varepsilon_v^2)$; $Z = X^T Z_1$; $J = X^T \text{diag}(3\varepsilon^2 - \kappa) X$; $\beta \leftarrow \beta - J^{-1} Z$. The iterations continue until the prescribed convergence criterion is satisfied: $\|Z\|_2 < 10^{-6}$. In practice, convergence is typically achieved within 3-5 iterations. To ensure numerical stability, if the specified divergence condition is met, $\|\beta - \hat{\beta}_{OLS}\| > 10$ the algorithm returns the initial estimate $\hat{\beta}_{OLS}$.

The theoretical efficiency coefficient of PMM3 is

$$g_3 = 1 - \frac{\gamma_4^2}{6 + 9\gamma_4 + \gamma_6}, \quad (1)$$

where $\gamma_4 = \mu_4/\mu_2^2 - 3$, $\gamma_6 = \mu_6/\mu_2^3 - 15(\mu_4/\mu_2^2) + 30$ – normalized fourth and sixth-order cumulants. An important advantage of PMM3 is that it does not require knowledge of the distribution: cumulants are estimated directly from the sample.

2.5 Asymptotic normality of PMM3

Under the standard regularity conditions, the PMM3 estimator is consistent and asymptotically normally distributed. $\sqrt{n}(\hat{\beta}_{PMM3} - \beta) \rightarrow^d N(0, g_3 \cdot \sigma^2 \cdot (X^T X / n)^{-1})$ Its asymptotic covariance matrix is reduced by a factor of $(1/g_3)$ relative to that of the OLS estimator: $\text{Var}(\hat{\beta}_{PMM3}) = g_3 \cdot \text{Var}(\hat{\beta}_{OLS})$

For the Gaussian case ($\gamma_4 = 0$) $g_3 = 1$ PMM3 is asymptotically equivalent to OLS. For platykurtic distributions ($\gamma_4 < 0$) $g_3 < 1$ PMM3 is more efficient than OLS. The consistency of PMM3 depends only on the assumption of zero-mean errors and does not require specification of the exact distributional form. A more detailed derivation is provided in [10] and [20]. The three methods differ in what they must assume. OLS and PMM3 both estimate only the p coefficients of the vector and require no knowledge of the distributional form, whereas MLE-TN estimates $p + 2$ parameters and requires the TN form to be correctly specified. Under TN errors OLS is the baseline ($ARE = 1$), PMM3 attains $g_3 < 1$ whenever the fourth-order cumulant is negative, and MLE-TN attains the Cramér–Rao bound.

2.6 Theoretical characteristics of the TN distribution

The shapes of the TN density are shown in the left panel of Fig. 1 and the theoretical efficiency curve $g_3(\lambda)$ in the right panel. Evaluated at the six shape values used throughout this study, the moment formulas give $\gamma_4 = -0.080, -0.500, -0.959, -1.280, -1.620$ and -1.849 with $g_3 = 0.9988, 0.9286, 0.6574, 0.3868, 0.1490$ and 0.0454 for $\lambda = 0.5, 1.0, 1.5, 2.0, 3.0$ and 5.0 , corresponding to a variance reduction relative to OLS of

0.1%, 7.1%, 34.3%, 61.3%, 85.1% and 95.5%. The distribution is bimodal for $\lambda > 1$, that is for every $\lambda \geq 1.5$ in this set.

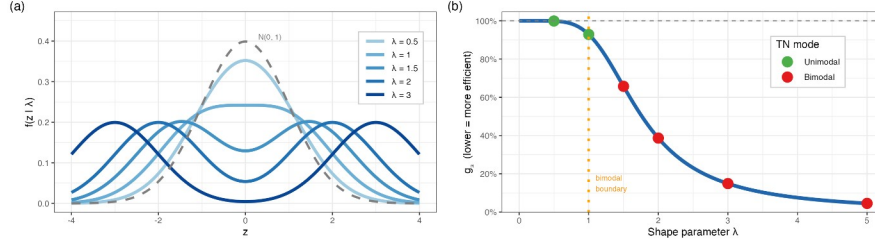


Fig. 1. Two-piece Normal model. Left: TN densities for $\lambda \in \{0.5, 1, 1.5, 2, 3\}$, standard normal dashed. Right: theoretical efficiency g_3 as a function of λ ; the dotted line marks the bimodality boundary $\lambda = 1$, unimodal cases green, bimodal red.

3 Results and discussion

3.1 Monte Carlo study

The data-generating model is defined as:

$$y_v = 1 + \beta_1 x_v + \varepsilon_v, \quad \varepsilon_v \sim TN(0, 1, \lambda). \quad (1)$$

The simulation parameters: number of replications $M=500$; sample sizes $n \in \{46, 100, 200, 500\}$; shape parameters $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$; random seed = 20260312. Metrics: Bias= $\hat{\beta}_1 - 1$; Var($\hat{\beta}_1$); MSE=Bias²+Var; ARE=Var(OLS)/Var(method); $g_{3_emp} = \text{Var(PMM3)}/\text{Var(OLS)}$.

The results for all 24 scenarios are summarised in Figs. 2 and 3; the full numeric table is provided in the supplementary material (see Data and Code Availability).

Here ARE > 1 indicates superiority over OLS. Furthermore $g_{3_emp} = \text{Var(PMM3)}/\text{Var(OLS)}$, $\Delta g_3 = (g_{3_emp} - g_{3_theor})/g_{3_theor} \times 100\%$. For $n \geq 100$ the deviation remains within $\pm 10\%$.

PMM3 significantly outperforms OLS at $\lambda \geq 1.5$: ARE ranging from 1.5 to 23.8 – the theory has been confirmed. MLE-TN is even more effective than PMM3 for $\lambda \geq 1.5$ (assuming the TN specification is correct), but it estimates $p+2$ parameters – excessive parameterization for small n . The empirical efficiency coefficient g_{3_emp} approximates its theoretical value g_{3_theor} with an accuracy of approximately $\pm 10\%$ for sufficient sample sizes ($n \geq 100$). For the most challenging scenario, $\lambda=5$, $n=100$, the efficiencies were ARE(PMM3) = 23.8 compared with ARE(MLE-TN) = 27.4. Thus, PMM3 lags behind MLE-TN by only about 15%, while requiring no prior knowledge of the underlying error distribution.

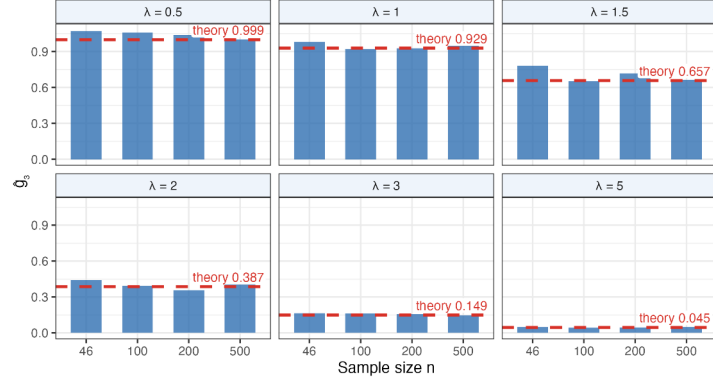


Fig. 2. Empirical efficiency coefficient $\hat{g}_3$ (bars) against its theoretical value (dashed line) for all 24 Monte Carlo scenarios (six values of $\lambda \times$ four sample sizes n , $M = 500$).

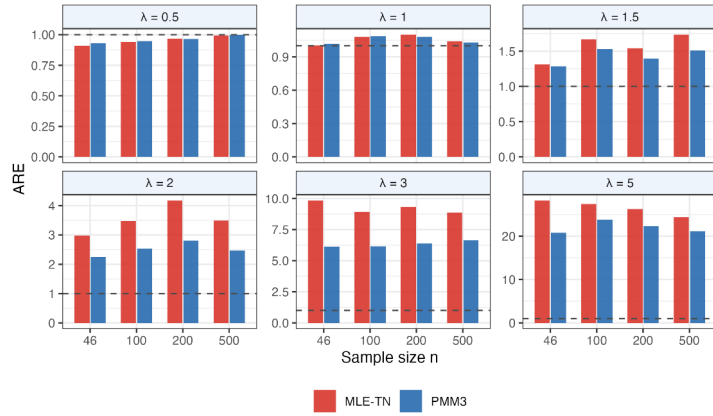


Fig. 3. Monte Carlo ARE of MLE-TN and PMM3 relative to OLS, for six shape values λ and four sample sizes n ($M = 500$). ARE > 1 means more efficient than OLS.

3.2 Real-world data: Fisher's iris dataset (Iris versicolor)

A subset of the classical Fisher Iris dataset [3] corresponding to the species *Iris versicolor* was used for empirical evaluation. The dataset contains $n = 50$ observations.

$$Sepal.Length_v = \beta_0 + \beta_1 \cdot Sepal.Width_v + \varepsilon_v. \quad (1)$$

OLS Residual Diagnostics: $n = 50$; $\hat{\gamma}_3 = 0.273$ (indicating only negligible asymmetry); $\hat{\gamma}_4 = -0.966$ (indicates pronounced platykurticity); $span / SD \approx 3.58$ – suggesting a restricted measurement range.

The corresponding theoretical efficiency is: $g_{3theor} = 0.483$ which corresponds to a 51.7% reduction in variance. Based on the residual diagnostics: $\gamma_4 = -0.966 < -0.7$ PMM3 is recommended.

Table 1. Comparison of estimation methods for iris versicolor ($n = 50$).

Method	$\hat{\beta}_1$	LOO-MSE	g_{3_emp} (bootstrap)
OLS	0.865	0.2070	1.000 (baseline)
PMM3	0.797	0.2022	0.570
MLE-TN	0.753	0.2109	0.746

The relative variance is computed as $g_{3_emp} = \text{Var}(\text{method})/\text{Var}(\text{OLS})$ using bootstrap estimation with $B = 2000$ replications.

PMM3 achieved the smallest prediction error: LOO-MSE = 0.2022, which is lower than that obtained by OLS = 0.2070 and MLE-TN = 0.2109. Bootstrap analysis indicates a 43% reduction in variance ($g_{3_emp} = 0.570$), compared with the theoretical prediction of 51.7%. Given the relatively small sample size ($n = 50$), this agreement can be considered satisfactory. Despite the additional parameters (η, λ) , MLE-TN exhibited the poorest performance. Consequently, for datasets characterized by $n = 50$ and $\gamma_4 \approx -0.97$, PMM3 represents the optimal estimation strategy.

Negative example. For the whiteside dataset (MASS package, household gas consumption versus outdoor temperature, $n = 56$) the residuals give $\gamma_4 = -0.431$, i.e. above the threshold: the theoretical gain is only 8.0% ($g_{3_theor} = 0.920$), the bootstrap ratio is $g_{3_emp} = 0.978$ and LOO-MSE PMM3/OLS = 0.999. Five further datasets with $\gamma_4 \in [-0.54, -0.12]$ give $g_{3_emp} \in [1.03, 1.36]$ and LOO-MSE PMM3/OLS ≤ 1.013 , i.e. no gain, which supports the threshold $\gamma_4 \approx -0.7$.

3.3 Robustness under model misspecification

Robustness was assessed by generating data from non-TN error distributions (Table 2), with sample size $n = 100$, number of Monte Carlo replications $M = 500$.

Table 1. ARE under model misspecification ($n = 100, M = 500$).

True DGP	γ_4 (true)	ARE(MLE-TN)	ARE(PMM3)
Uniform	-1.200	1.655	2.685
Triangular	-0.600	0.969	1.061
Logistic	+1.200	0.994	1.023
Student's t10	+1.000	0.982	1.047

PMM3 is robust: ARE ≥ 1.02 for all four distributions, so it performs no worse than OLS even when its assumptions fail; for the uniform law ($\gamma_4 = -1.2$) it reaches 2.685, above MLE-TN (1.655). MLE-TN is considerably more sensitive to misspecification:

under the triangular law its ARE = 0.969 is below OLS. The Newton–Raphson implementation converged in 100% of the 200 replications with no NA outcomes for every $\lambda \in \{0.8, 1.0, 1.2, 1.5, 2.0, 3.0\}$, including the strongly bimodal cases, in 2.8–3.7 iterations on average, with absolute bias below 0.003.

4 Conclusions

1. Theoretical results. PMM3 provides a 7% to 95% reduction in variance relative to OLS for TN-distributed errors with $\lambda \in \{1, \dots, 5\}$. Under correct model specification, MLE-TN can achieve an even greater efficiency gain; however, it requires estimation of $p + 2$ parameters. Monte Carlo study. MLE-TN is substantially more efficient than PMM3 for large values of λ , provided that the TN specification is correct. The empirical efficiency coefficient g_3 of PMM3 converges to its theoretical value for sample sizes of approximately $n \geq 100 - 200$.
2. Robustness. PMM3 never falls below OLS under model misspecification (ARE from 1.02 to 2.68), whereas MLE-TN may perform worse than OLS. PMM3 is therefore the safe choice when the error distribution is uncertain.
3. Real-world data. For the Iris versicolor dataset ($\gamma_4 = -0.966$, $n = 50$) PMM3 achieves a 43% reduction in variance, while its predictive performance satisfies LOO-MSE $\text{PMM3} < \text{OLS} < \text{MLE-TN}$.
4. Empirical threshold. PMM3 demonstrates a practical advantage over OLS only when $\gamma_4 \lesssim -0.7$. For $-0.7 < \gamma_4 < 0$ no method offers a practical advantage over OLS (prediction error within 2% of OLS). In practice: prefer MLE-TN when the TN model is justified and n is large, PMM3 when the error distribution is uncertain, and OLS for approximately Gaussian errors.

Data and Code Availability. The R code of the study, the stored results of all experiments and a script that reruns the simulation and checks it against the stored tables are openly available under the MIT licence at <https://github.com/SZabolotnii/Ku-PMM3-MLE-TN-code-supplement> and archived on Zenodo: <https://doi.org/10.5281/zenodo.22849684>.

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